

EGM 3601 Solid Mechanics

Bonus Project 1

Stress Analysis of a Cantilever Beam

Problem Specification

Consider the beam in the figure below. It is clamped on the left side and has a point force of 8kN acting downward on the right end of the beam. The beam has a length of 4 meters, width of 0.346 meters and height of 0.346 meters (cross-section is a square). Additionally, the beam is composed of a material which has a modulus of elasticity (E) of 200GPa, poisson's ratio of 0.3 and shear modulus (G) of 76.92 GPa.

Using ANSYS APDL, please complete the following steps:

1. Simulate the **maximum stress in the beam**.
2. Compare your simulation result to the analytical solution and calculate the percent error. Upload your calculations and results as a PDF file to webcourses.
3. Upload the video file of your ANSYS simulation to the Webcourses. **Please show the steps and the results of the simulation on your video.**

